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MZ 505
Inventory Systems

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Session 07:
The Newsvendor Model

Adapted from Cachon/Terwisch (2013)

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Newsvendor basic characteristics

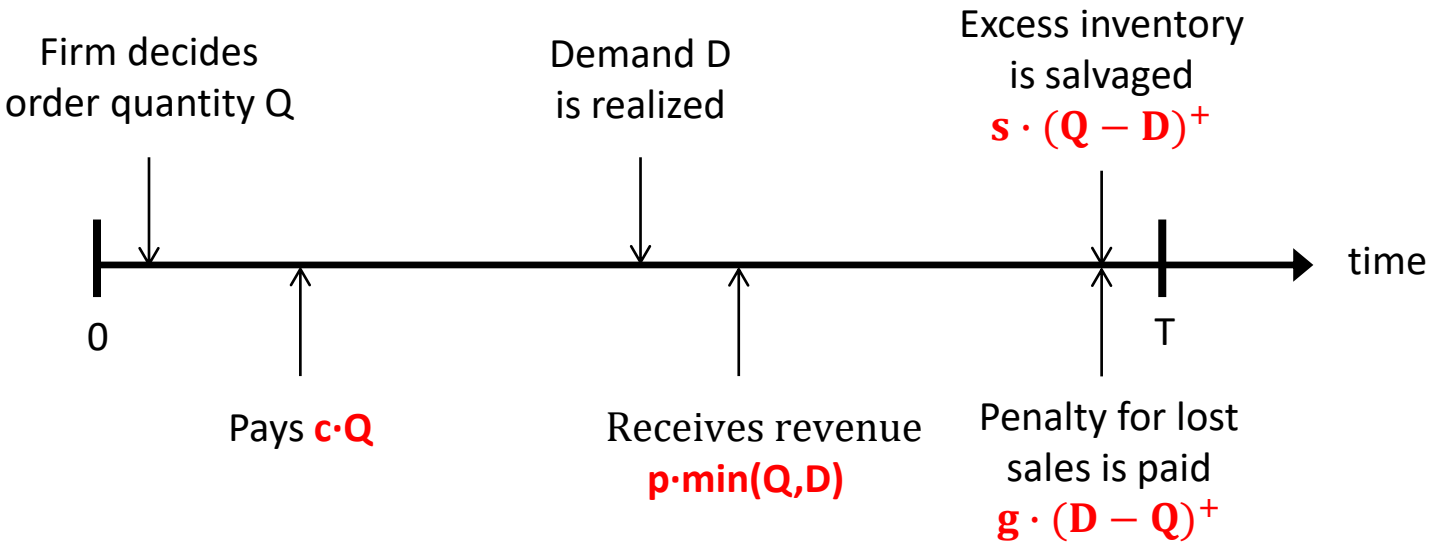
- Random demand.
- Product has a single selling season.
  - Newspaper, fresh consumable products, etc.
  - Increasingly many major non-consumable items are falling into the style goods category.
  - Books, toys, fashion garments, laptops, tablets.
- Long lead time - a single opportunity to procure (or produce) at the start of the season.
  - Production occurs in “Low Cost Country” for several of these products.
- Limited salvage value at the end of the season.

Newsvendor model implementation steps

- Generate a demand model:
  - Determine a distribution function that accurately reflects the possible demand outcomes; e.g., Normal distribution function, Poisson, etc.
- Gather economic inputs:
  - Selling price, production/procurement cost, salvage value of inventory, goodwill cost.
- Choose an objective:
  - e.g. maximize expected profit or satisfy a service level target.
- Choose a quantity to order.

The Newsvendor Model: Basic setting

- The problem parameters are as follows:
  - D: demand (a random variable)
  - Q: order quantity
  - p: unit revenue
  - c: unit cost
  - s: salvage value per unit leftover
  - g: goodwill cost per unit short



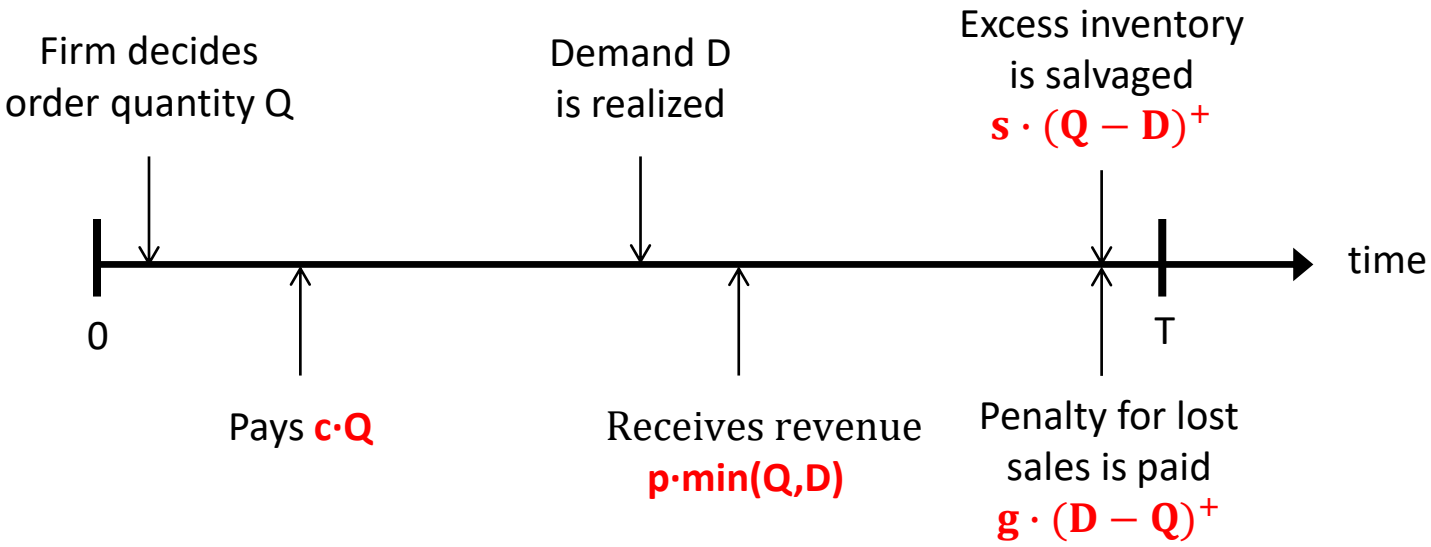
$(a)^+ = \max(0, a)$

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Newsvendor model: Objective function

- Objective function: choose Q to maximize the total expected profit

$$\pi(Q) = pE[Sold] + sE[Leftover] - gE[Short] - cQ$$



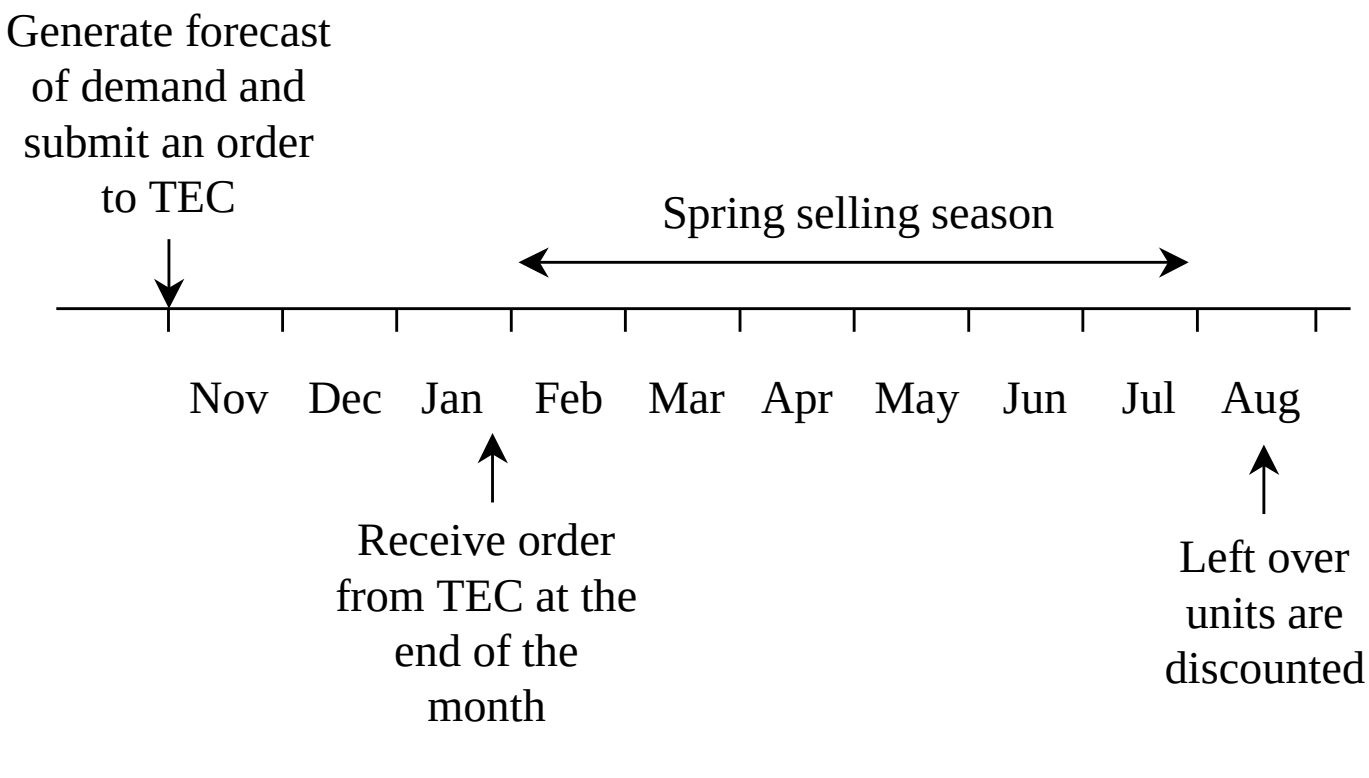
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O'Neill's Hammer 3/2 wetsuit





Hammer 3/2 timeline





Hammer 3/2 economics

- The demand model is a normal distribution with mean 3192 and standard deviation 1181.
- O'Neill sells each suit for p = $190
- O'Neill purchases each suit from its supplier for c = $110 per suit.
- Discounted suits sell for s = $90
  - This is also called the salvage value.
- O'Neill has the **"too much/too little problem"**:
  - Order too much and inventory is left over at the end of the season.
  - Order too little and sales are lost.



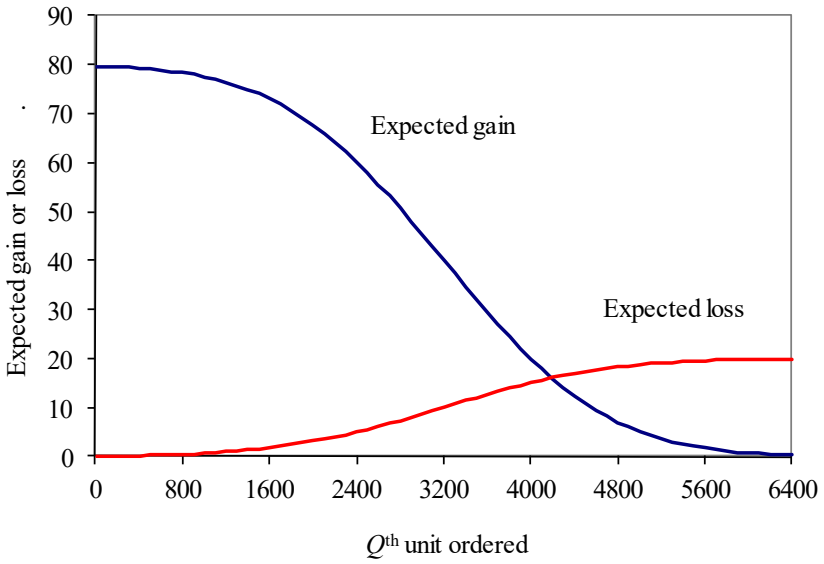
"Too much" and "too little" costs

O'Neill: p=190, c = 110, s = 90, g = 0

- **C_o = overage cost**
  - The cost of ordering one more unit than what you would have ordered had you known the demand.
  - Suppose you had left over inventory; C_o is the increase in profit you would have realized had you ordered one unit less.
  - For this example, C_o = Cost – Salvage value = c – s = $20
- **C_u = underage cost**
  - The cost of ordering one unit less than what you would have ordered had you known the demand.
  - Suppose you had lost sales; C_u is the increase in profit you would have realized had you ordered one more unit.
  - For this example, C_u = Price – Cost + Goodwill = p – c + g = $80

Balancing the risk and benefit of ordering a unit

- Ordering one more unit increases the chance of overage …
  - Expected loss on the Qth unit $= C_o \times F(Q)$
  - $F(Q) = Pr(D \leq Q)$ → CDF of demand distribution
- But, ordering one more unit decreases the chance of underage…
  - Expected gain on the Qth unit $= C_u \times (1 - F(Q))$



As more units are ordered, the expected benefit from ordering one unit decreases while the expected loss of ordering one more unit increases.

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The critical ratio and optimal order quantity

- To maximize expected profit, order Q units so that the expected loss on the Qth unit equals the expected gain on the Qth unit:

$$C_o \times F(Q) = C_u \times (1 - F(Q))$$

- Rearranging terms in the above equation

$$F(Q^*) = Pr(D \leq Q^*) = \frac{C_u}{C_u + C_o} = \frac{p - c + g}{p - s + g} = \alpha$$

- For the Hammer 3/2 the critical ratio is

$$\frac{C_u}{C_u + C_o} = \frac{80}{80 + 20} = 0.8$$

- Using the Standard Normal distribution table (with the round-up rule) , $z(0.8) = ?$

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The Standard Normal Distribution Function Table

Values of z on the outside

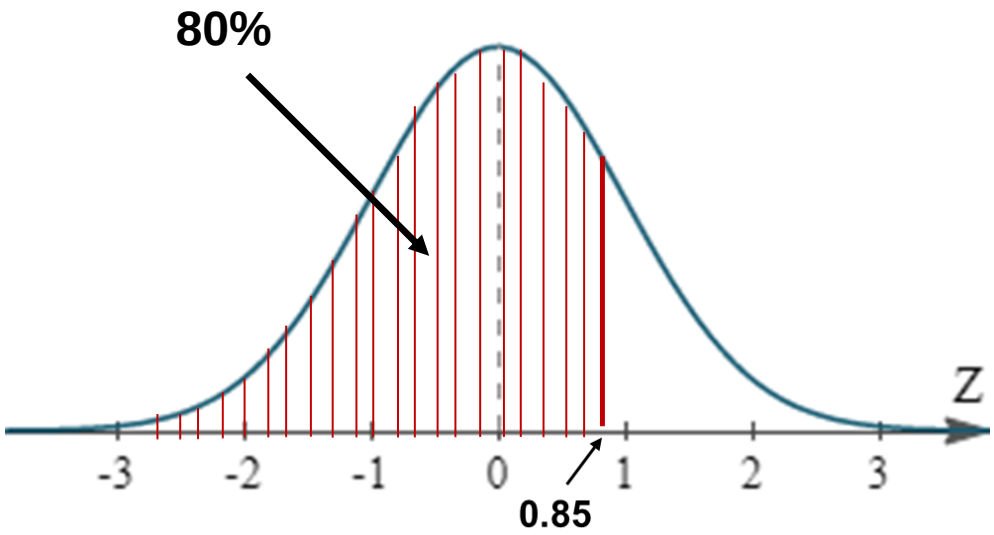
Standard Normal Distribution Function Table (continued), $\Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621

Probabilities in the inside

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The critical ratio and optimal order quantity



- Using the standard normal distribution table (with the round-up rule) , $z(0.8) = 0.85$.

- Then, the optimal quantity is equal to:

$$Q^* = \mu + z(\alpha) \cdot \sigma = 3{,}192 + 0.85 \cdot 1{,}181 = 4{,}196$$

The Newsvendor Model: Performance Measures

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Performance Measures

- For any order quantity we would like to evaluate the following performance measures:
  - ***In-stock probability* (*Type I - service level*)**
    - Probability all demand is satisfied.
  - ***Stockout probability***
    - Probability some demand is lost.
  - ***Fill rate* (*Type II - service level*)**
    - The fraction of demand that is filled.
  - ***Expected lost sales***
    - The expected number of units by which demand will exceed the order quantity.
  - ***Expected sales***
    - The expected number of units sold.
  - ***Expected left over inventory***
    - The expected number of units left over after demand is realized.
  - ***Expected profit***

In-stock probability

- The in-stock probability is the probability all demand is satisfied
  - All demand is satisfied if $D \leq Q$
  - Hence, if we stock optimally, the **in-stock probability** is $F(Q) = \Phi(z)$
  - It follows that the **stockout probability** is $1 - F(Q) = 1 - \Phi(z)$
- **Question**: In the Hammer 3/2 example, **what is the in-stock probability if the order quantity is Q = 3000?**
  - Evaluate the z-statistic for the order quantity:

$$z = \frac{Q-\mu}{\sigma} = \frac{3000-3192}{1181} = -0.16$$

  - Look up $\Phi(z)$ in the Standard Normal Distribution Function Table: $\Phi(-0.16) = ?$

The Standard Normal Distribution Function Table

Standard Normal Distribution Function Table (continued), $\Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621

$$\Phi(-0.16) = 1 - \Phi(0.16) = 1 - 0.5636 = 0.4364$$

- **Answer**: If 3000 units are orders, there is a 43.64% chance all demand will be satisfied.

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Choose Q subject to a minimum in-stock probability

- Suppose we wish to find the **order quantity** for the Hammer 3/2 **that generates at least a 99% in-stock probability**.
- **Step 1**:
  - Find the z-statistic that yields the target in-stock probability.

The Standard Normal Distribution Function Table

Standard Normal Distribution Function Table, $\Phi(z)$ (Concluded)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
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0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936

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Choose Q subject to a minimum in-stock probability

- Suppose we wish to find the **order quantity** for the Hammer 3/2 **that generates at least a 99% in-stock probability**
- **Step 1**:
  - Find the z-statistic that yields the target in-stock probability.
  - In the Standard Normal Distribution Function Table we find $\Phi(2.32) = 0.9898$ and $\Phi(2.33) = 0.9901$
  - Choose $z(0.99) = 2.33$ to meet in-stock probability target.
- **Step 2**:
  - Convert the z-statistic into an order quantity for the actual demand distribution.
  - $Q = \mu + z(\alpha) \cdot \sigma = 3{,}192 + 2.33 \cdot 1{,}181 = 5{,}944$

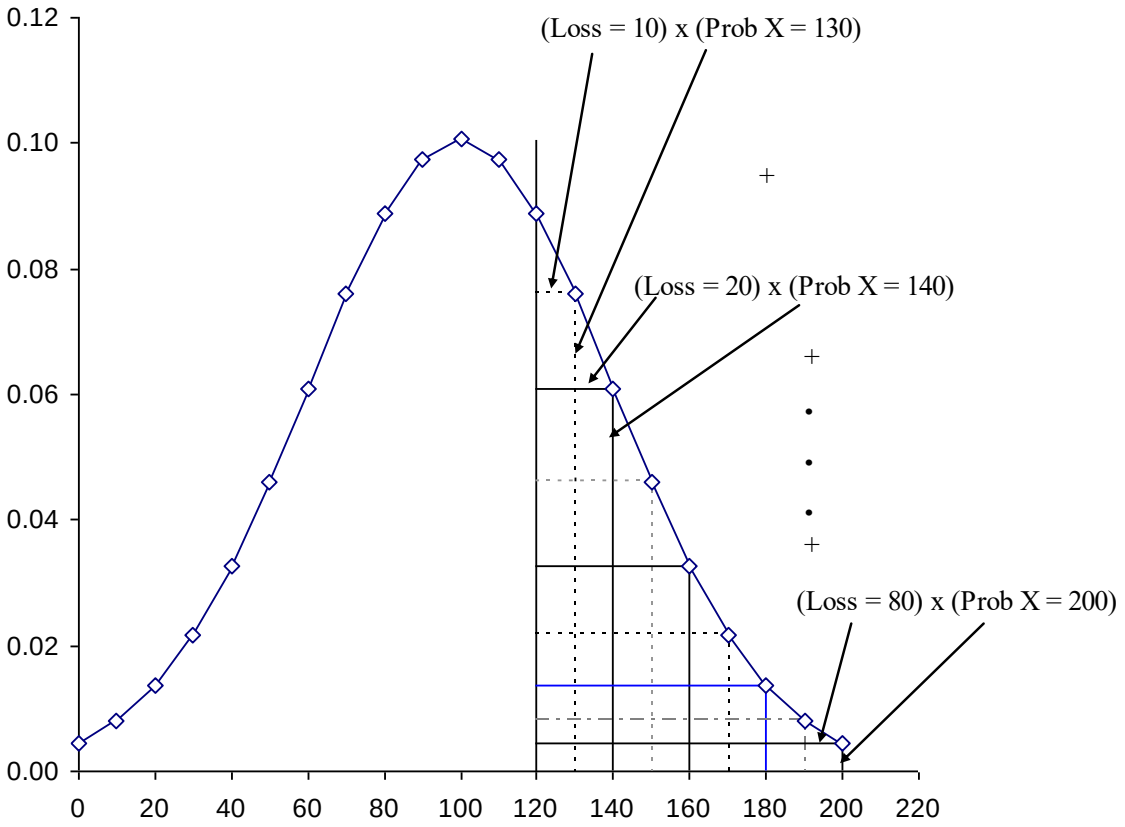
Expected Lost Sales (Expected Shortage)

- So far, our only performance measure was the "Type I - service level" or "in-stock probability".
- Production/inventory quantity was chosen to achieve "optimal" service which maximizes profit (or a target in-stock probability).
- We may, however, also be interested in other performance measures, e.g., what is the **expected lost sales** (or shortage) if we order $Q^* = F^{-1}(a) = \mu + z(a)\sigma$??
- E[short] is the expected shortage in a season, e.g., how many wetsuits we ordered too little.
- E[short] is not a percentage, it is the number of units:

$$E[\text{short}] = E[(D - Q)^+] = \int_Q^\infty (D - Q) f(D) dD$$

Expected lost sales: graphical explanation

- Suppose demand can only be one of the following values {0,10,20,…,190,200}
- Suppose Q = 120
- Let d = actual demand
- What is the expected lost sales?
- If d ≤ Q, lost sales = 0
- If d = 130, lost sales = d – Q = 10
- Expected lost sales = 10 x Prob{D = 130} + 20 x Prob{D = 140} +… + 80 x Prob{D = 200}



Normal Loss Function

- Standard Normal Loss Function: $L(z) = \varphi(z) - z \times (1 - \Phi(z))$

  → Standard Normal Loss Function Table

- If D follows a normal distribution with (μ, σ) and the order quantity is Q, then:

  A. Evaluate the z-statistic for Q: $z = \frac{Q-\mu}{\sigma}$

  B. Use the z-statistic to look up $L(z)$ in the Standard Normal Loss Function Table.

     Excel: $L(z) = \text{Norm.s.dist}(z, 0) - z \times (1 - \text{Norm.s.dist}(z, 1))$

  C. Compute **expected lost sales**: $E[\text{short}] = \sigma \times L(z)$

❖ The steps for calculating ESC when demand follows a Poisson distribution are outlined at the end of this slideshow

Expected lost sales of Hammer 3/2s with Q = 4196*

- Suppose O'Neill orders 4196 Hammer 3/2s (optimal quantity)
- **What will be O'Neill's lost sales (in units) on average?**
- To find the answer:
  - **Step 1**: normalize the order quantity to find its z-statistic

$$z = \frac{Q-\mu}{\sigma} = \frac{4196-3192}{1181} = 0.85$$

  - **Step 2**: Look up in the Standard Normal Loss Function Table the expected loss for a standard normal distribution with that z-statistic: $L(0.85) = ?$

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The Standard Normal Loss Function Table

Standard Normal Loss Function Table, $L(z)$ (Concluded)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.3989	0.3940	0.3890	0.3841	0.3793	0.3744	0.3697	0.3649	0.3602	0.3556
0.1	0.3509	0.3464	0.3418	0.3373	0.3328	0.3284	0.3240	0.3197	0.3154	0.3111
0.2	0.3069	0.3027	0.2986	0.2944	0.2904	0.2863	0.2824	0.2784	0.2745	0.2706
0.3	0.2668	0.2630	0.2592	0.2555	0.2518	0.2481	0.2445	0.2409	0.2374	0.2339
0.4	0.2304	0.2270	0.2236	0.2203	0.2169	0.2137	0.2104	0.2072	0.2040	0.2009
0.5	0.1978	0.1947	0.1917	0.1887	0.1857	0.1828	0.1799	0.1771	0.1742	0.1714
0.6	0.1687	0.1659	0.1633	0.1606	0.1580	0.1554	0.1528	0.1503	0.1478	0.1453
0.7	0.1429	0.1405	0.1381	0.1358	0.1334	0.1312	0.1289	0.1267	0.1245	0.1223
0.8	0.1202	0.1181	0.1160	0.1140	0.1120	0.1100	0.1080	0.1061	0.1042	0.1023
0.9	0.1004	0.0986	0.0968	0.0950	0.0933	0.0916	0.0899	0.0882	0.0865	0.0849
1.0	0.0833	0.0817	0.0802	0.0787	0.0772	0.0757	0.0742	0.0728	0.0714	0.0700
1.1	0.0686	0.0673	0.0659	0.0646	0.0634	0.0621	0.0609	0.0596	0.0584	0.0573

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Expected lost sales of Hammer 3/2s with Q = 4196*

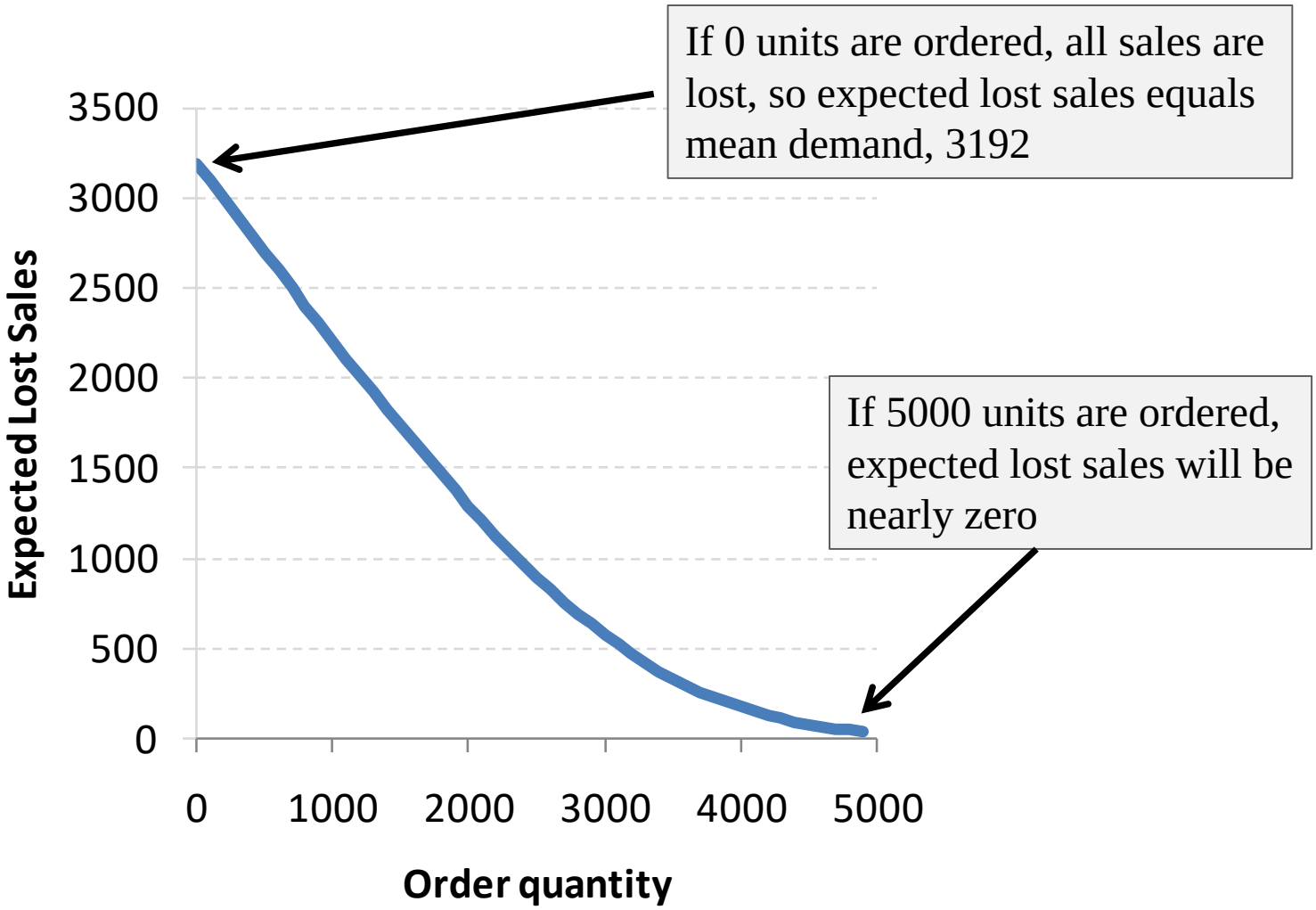
- Suppose O'Neill orders 4196 Hammer 3/2s (optimal quantity)
- **What will be O'Neill's lost sales (in units) on average?**
- To find the answer:
  - **Step 1**: normalize the order quantity to find its z-statistic

$$z = \frac{Q-\mu}{\sigma} = \frac{4196-3192}{1181} = 0.85$$

  - **Step 2**: Look up in the Standard Normal Loss Function Table the expected loss for a standard normal distribution with that z-statistic: $L(0.85) = 0.1100$
  - **Step 3**: Evaluate lost sales for the actual normal distribution:

$$E[\text{short}] = \sigma \times L(z) = 1181 \times 0.11 = 130 \text{ units}$$

Expected lost sales of Hammer 3/2s



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Measures that follow expected lost sales

- **Expected sales**
  - $E[\text{sales}] = E[\min(D, Q)] = E[D - (D - Q)^+] = \mu - E[\text{short}]$
- **Expected leftover inventory**
  - $E[\text{left}] = Q - E[\text{sales}] = Q - \mu + E[\text{short}]$
- **Expected profit**
  - $E[\text{profit}] = \pi(Q) = pE[\text{sales}] + sE[\text{left}] - gE[\text{short}] - cQ$

❖ **Note**: the above equations hold for <u>any demand distribution</u>

Question: what is the expected profit if they order 4196 Hammer 3/2s?

Expected profit if we order 4196 units of Hammer 3/2s

- **Expected lost sales**
  - $E[short] = \sigma L(z) = 1181 \times 0.11 = 130$ units (for normal D)
- **Expected sales**
  - $E[sales] = \mu - E[short] = 3192 - 130 = 3062$ units
- **Expected leftover inventory**
  - $E[left] = Q - E[sales] = 4196 - 3062 = 1134$ units
- **Expected profit**
  - $E[profit] = \pi(Q) = pE[sales] + sE[left] - gE[short] - cQ$

$$= 190 \cdot 3062 + 90 \cdot 1134 - 0 \cdot 130 - 110 \cdot 4196$$

$$= \$\mathbf{222,280}$$

Service Level vs. Fill Rate

- **Fill Rate** (or expected fill rate), which often is also called "Type II - service level", is another important measure.
- Fill rate represents the fraction of demand that is filled.
  - When we measure it ex-ante (i.e., we do not yet know demand), we call it "expected fill rate" and it is equal to the average fraction of demand that is filled

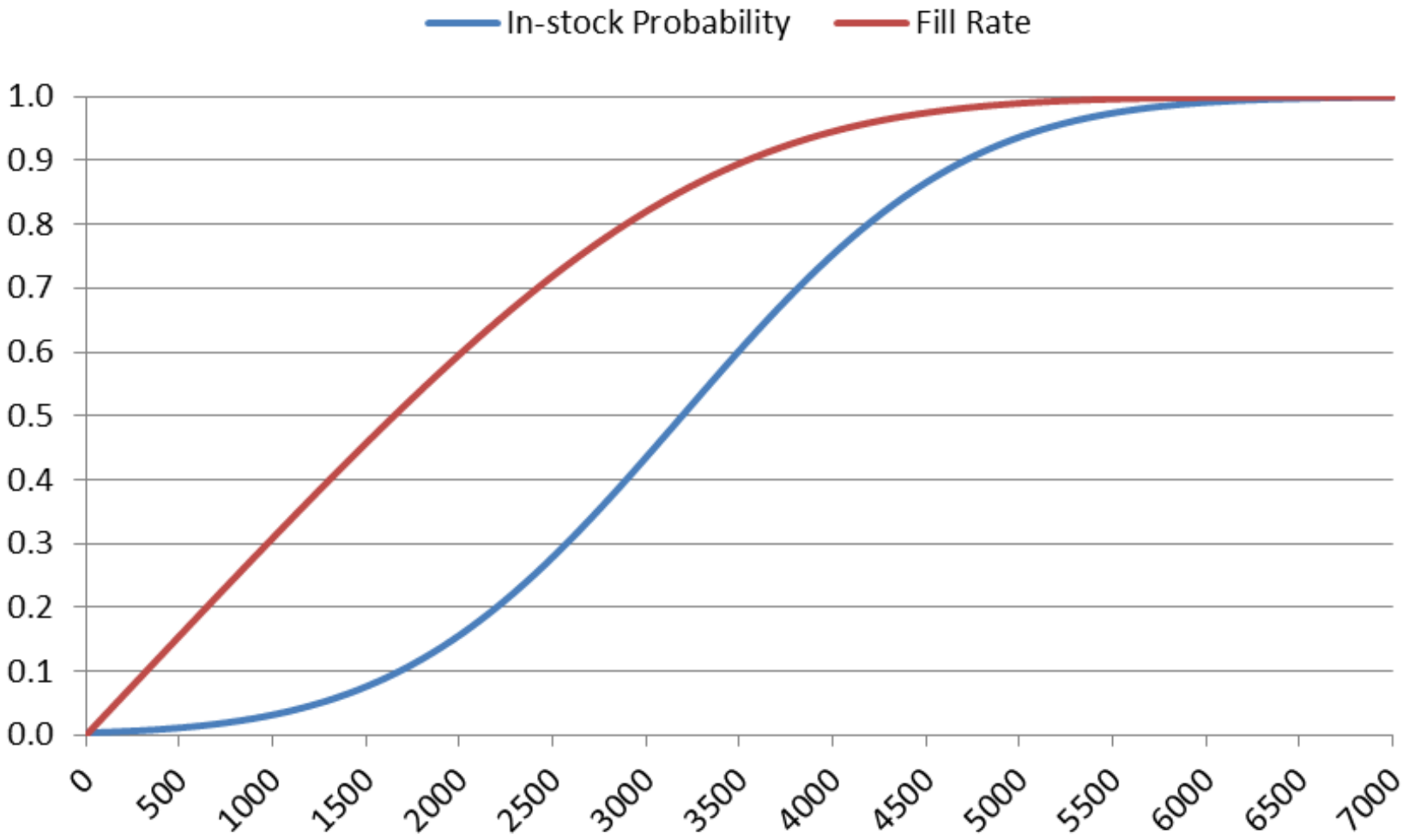
$$\text{Expected Fill Rate} = \frac{\text{E[sales]}}{\mu} = 1 - \frac{\text{E[short]}}{\mu}$$

- For the Hammer 3/2s case when we order 4196 units

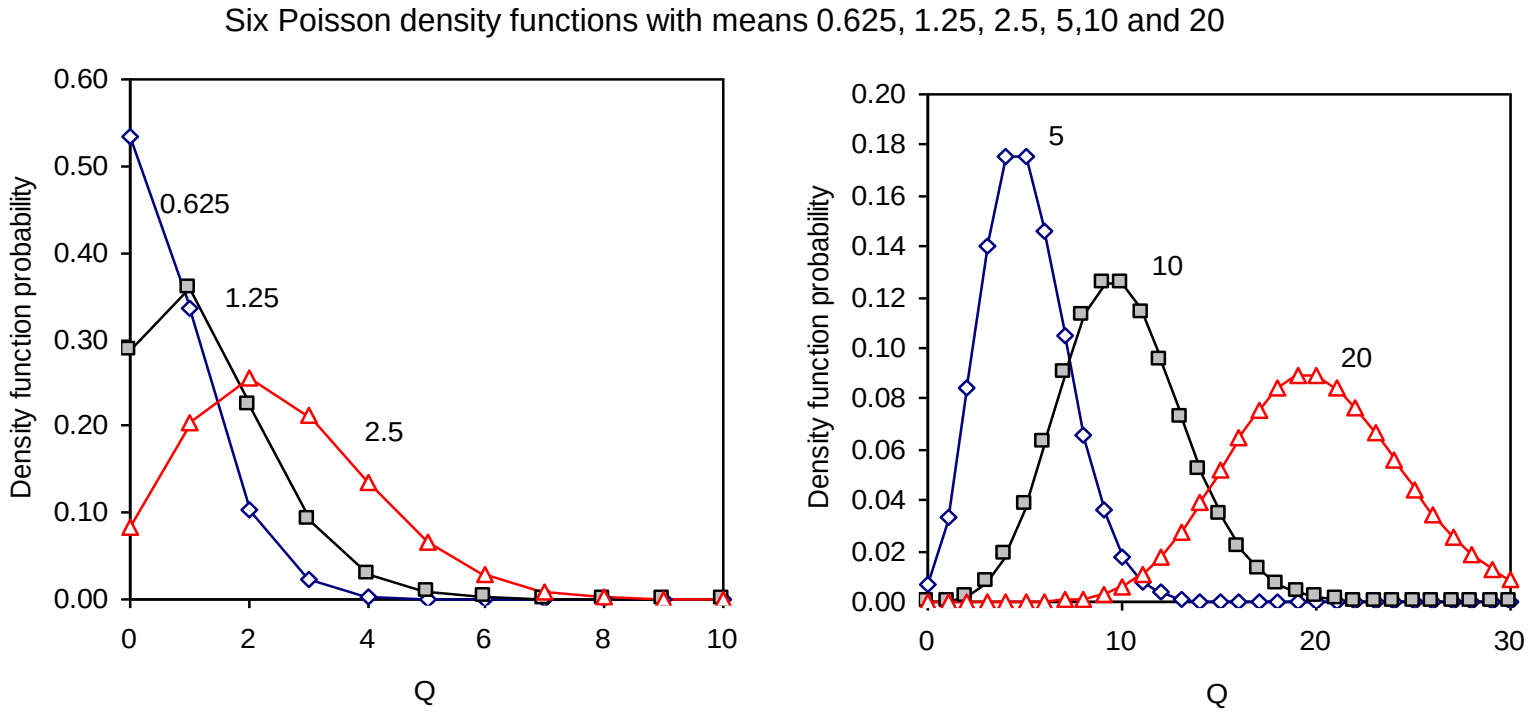
$$\text{Expected Fill Rate} = \frac{\text{E[sales]}}{\mu} = \frac{3062}{3192} = 0.96 \text{ or } 96\%$$

- Observe that expected fill rate is higher than the service level (in-stock probability of 80%). Why?

Service level vs. fill rate in Hammer 3/2s case



Poisson distribution



- Discrete distribution function: only non-negative integers
- Defined only by its mean, often denoted by λ
- Its variance is equal to the mean (hence $\sigma = \sqrt{\lambda}$)
- Good for modeling demands with low means (e.g., less than 20)

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Poisson distribution (continued)

- If the inter-arrival times of customers are exponentially distributed, then the number of customers that arrive in a given interval of time has a Poisson distribution (and vice versa).
- Poisson Mass Function:

$$P(D = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

- Poisson Cumulative Distribution Function:

$$F(Q) = P(D \leq Q) = \sum_{x=0}^{Q} P(x) = \sum_{x=0}^{Q} \frac{\lambda^x e^{-\lambda}}{x!}$$

Poisson distribution: Example

- TeaWorld plans to sell a Halloween thematic "tea set" in October.
- The set's price is $80, purchase cost is $35 and after the end of October any left over inventory will be cleared out at $20.
- Estimated demand at one of its stores is Poisson with mean $\lambda = 4.5$
- If only one order can be made for the season, what is the optimal order quantity?
  - ✓ C_o=35-20=15; C_u=80-35=45; $C_u / (C_u + C_p) = 45 / 60 = 0.75$

poisson.dist
(x; λ; false)

x	P(D = x)	P(D ≤ x)
0	0.0111	0.0111
1	0.0500	0.0611
2	0.1125	0.1736
3	0.1687	0.3423
4	0.1898	0.5321
5	0.1708	0.7029
6	0.1281	0.8311
7	0.0824	0.9134
8	0.0463	0.9597

poisson.dist
(x; λ; true)

Optimal quantity

Poisson Loss Function

- As with the Standard Normal, we have also tables for the Poisson Distribution Function and the Poisson Distribution Loss Function.
- Consider a Poisson distribution with mean (rate) λ
  - Probability distribution function: $P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$
  - Cumulative distribution function F(Q)

$$F(Q) = \sum_{x=0}^{Q} P(x) = \sum_{x=0}^{Q} \frac{\lambda^x e^{-\lambda}}{x!}$$

Excel: F(Q) = POISSON.DIST(Q, λ, TRUE)

- Then, the expected loss function is given by:

$$E[Lost] = \lambda - \lambda \cdot F(Q-1) - Q \cdot (1 - F(Q))$$

Readings

- Chapter 12 (sections 12.5, 12.6, and 12.7) from Cachon & Terwiesch, 3rd edition

Thank you!